

Yash Panchal

US Citizen | (404) 933-1936 | yashcpanchal41@gmail.com | [linkedin.com/in/yashcpanchal](https://www.linkedin.com/in/yashcpanchal) | yashcpanchal.com

EDUCATION

Georgia Institute of Technology

Atlanta, GA

Bachelor of Science in Computer Science (Systems & Architecture and Intelligence Threads)

Aug. 2024 – May 2028

- **GPA:** 3.94 / 4.0 | Faculty Honors, Dean's List
- **Relevant Coursework:** Data Structures & Algorithms, Advanced Algorithms, Operating Systems, Computer Systems & Networks, Computer Organization, Object-Oriented Programming, Intro to AI
- **Activities:** AI@GT, Trading Club, Supercomputing Club, GT Experimental Rocketry, Brazilian Jiu Jitsu Club

EXPERIENCE

Visa

May 2026 – Aug. 2026

Software Engineering Intern

Foster City, CA

- Moved payments off premium secure hardware with a Java tokenization service, saving an estimated **\$9M/year**.
- Secured PII with mutual-TLS auth and keyed-hash cache lookups, isolating raw-card exposure to one service.
- Benchmarked and optimized the request path under sustained load, cutting average latency **59% to 49ms**.
- Architected stateless, event-driven workers over pooled backends, scaling to ~200 req/s per instance.
- Covered the service end to end with unit and integration tests, including per-field partial-failure cases.

Ancient Language Intelligence Lab

Jan. 2026 – Present

Undergraduate Researcher

Atlanta, GA

- Shipped an end-to-end pipeline detecting citations in raw Greek text, reaching **0.88 macro F1** across 3 classes.
- Built TEI/XML parsers aligning Greek to its translations by CTS, LLM-labeling a **23K-example** citation corpus.
- Fine-tuned Ancient Greek BERT with LoRA into a citation classifier, covering the 80% of untranslated texts.

Narb

Aug. 2025 – Dec. 2025

Software Engineering Intern

Remote

- Built a no-code web dev system where users build, edit, and deploy their web-apps, live for 10+ SMB clients.
- Engineered an AST symbol-graph RAG pipeline, cutting hallucinations **63%** against a flat-retrieval baseline.
- Automated CI/CD in LangGraph, gating Lighthouse and Web Vitals, going live in **<3 min** on **\$0.40** of tokens.

Georgia Tech Experimental Rocketry

Jan. 2025 – Present

Avionics Software Engineer

Atlanta, GA

- Reconfigured 6,000+ lines of embedded C into mission, manager, and driver tiers, removing 20+ cross-layer calls.
- Wrote real-time Kalman filter and state machine logic, porting the flight software to new avionics hardware.
- Added flight data logging and tests against real hardware, catching sensor faults before launch.

PROJECTS

IMC Prosperity Algorithmic Trading Competition | Top 0.5% team finish (71/14,000+)

Apr. 2026

- Built an event-driven backtester with a C++ price-time matching engine, replaying 20K+ order-book ticks/day.
- Optimized 12-parameter strategies in parallel with a cross-validated Optuna pipeline, lifting team profit **~46%**.

xv6 Kernel Extensions | C, x86 Assembly, QEMU, GDB

Dec. 2025

- Reworked virtual memory with copy-on-write fork and lazy page allocation, cutting fork() clock cycles **99%**.
- Built a user-space threading library with create/join, locks, and condition variables, avoiding kernel calls.
- Extended the file system with per-user permissions and a two-level index, scaling max file size 4MB to 4GB.

Jobly – AI-native job board | Python, FastAPI, MongoDB, Docker, Next.js

Aug. 2025

- Built a LangGraph + Playwright agent with async concurrency, parsing **1,000+** new URLs in 2 min.
- Scored resume-to-job fit from semantic vector search, skill, seniority, and location at **92%** user satisfaction.

re.mind – real-time dementia assistant | Kotlin, TensorFlow Lite, ML Kit, Gemini, Firebase

Feb. 2025

- Built an on-device TFLite vision pipeline recognizing known faces at **90%+ accuracy, sub-200ms** latency.
- Streamed AI-extracted insights to caretaker dashboards, adapting Gemini prompts across 3 dementia stages.

TECHNICAL SKILLS & INTERESTS

Languages: Python, Java, C, C++, Kotlin, TypeScript, JavaScript, SQL, Assembly, Bash, HTML/CSS

Frameworks: PyTorch, TensorFlow Lite, scikit-learn, LangGraph, FastAPI, React, Next.js, Vert.x

Developer Tools: Git, Docker, Kubernetes, AWS, Google Cloud, Linux, CI/CD, Firebase, PostgreSQL, Claude Code

Concepts: Data Structures & Algorithms, System Design, Distributed Systems, Concurrency, REST APIs, RAG

Interests: Competitive Strategy Games (Catan & Poker), Reading, Weightlifting, Running, Football, Hiking, Jiu Jitsu